Analytical Chemistry 1 (125101)

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Framework of this course

❖ Introduction (L1)

❖ Section I Chemical Measurements

1.1 Basic knowledge (L2)

1.2 Error and estimation (L3&L4)

1.3 Linear regression (L5)

❖ Section II Chemical Equilibrium

2.1 pH and buffer solutions (L6)

2.2 Titrations (L7&L8)

2.3 Potentiometry in titrations (L9)

2.4 Gran plot in titrations (L10)

2.5 Electroanalysis (L11)

2.6 gravimetric analysis (L12)

❖ Summary (L13)

Section II

Chemical Equilibrium

Targets

- **Principles of Chemical Equilibrium:** Equilibrium constant.
- **pH calculation:** strong/weak acid and base.
- **Buffer solution:** conjugate acids and bases, Henderson-Hasselbalch Equation, buffer capacity.

Principles of Chemical Equilibrium

Reaction: $a\,\mathbf{A} + b\,\mathbf{B} \underset{K'}{\overset{K}{\rightleftharpoons}} c\,\mathbf{C} + d\,\mathbf{D}$

Equilibrium constant: $$K = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

Standard unit of concentration:

Solutes: M (mol/L)

Gases: Atm.

Pure solid, liquid and solvent are omitted. e.g. [precipitate] = 1

[solvent] ~1

If the direction of a reaction is reversed, $K' = 1/K$

If two reactions are added, $K_3 = K_1 \times K_2$

Equilibrium constant of H_2O (K_w)

$$H_2O \overset{K_w}{\rightleftharpoons} H^+ + OH^-$$

w stand for water

$$K_w = [H^+][OH^-] \approx 10^{-14}$$ at 25.0 °C

Calculate the $[H^+]$ and $[OH^-]$ in pure water at 25.0 °C

$$H_2O \rightleftharpoons H^+ + OH^-$$

$$x\ M \quad x\ M$$

$$K_w = [H^+][OH^-] \approx 10^{-14}$$

$$x \times x = 10^{-14}$$

$$x = 1.0 \times 10^{-7}\ M$$

Test yourself find the $[H^+]$ in pure water at 0°C (K_w=1.15×10^{-15}).

$[H^+]$ is **3.4×10^{-8} M**

Equilibrium constant of H_2O (K_w)

Calculate the $[OH^-]$, if $[H^+] = 1.0\times10^{-3}$ M

(From now on, assume the T=25.0 °C unless otherwise stated)

$$K_w = [H^+][OH^-] = 10^{-14}$$

$$1.0\times10^{-3}\,[OH^-] = 10^{-14}$$

$$[OH^-] = 10^{-11}\text{M}$$

As the $[H^+]$ increases, the $[OH^-]$ necessarily decreases, vice versa, because the K_w is constant at a given T.

2.1 pH and buffer solutions

Overview

2.1-1 pH for different solutions

Strong/weak acids & bases in H_2O

2.1-2 Buffer solutions

Henderson – Hasselbalch equation

Buffer capacity, β

Terms to understand

Acid dissociation constant, K_a

Base dissociation constant, K_b

Conjugate acid-base pair

Fraction of dissociation, α

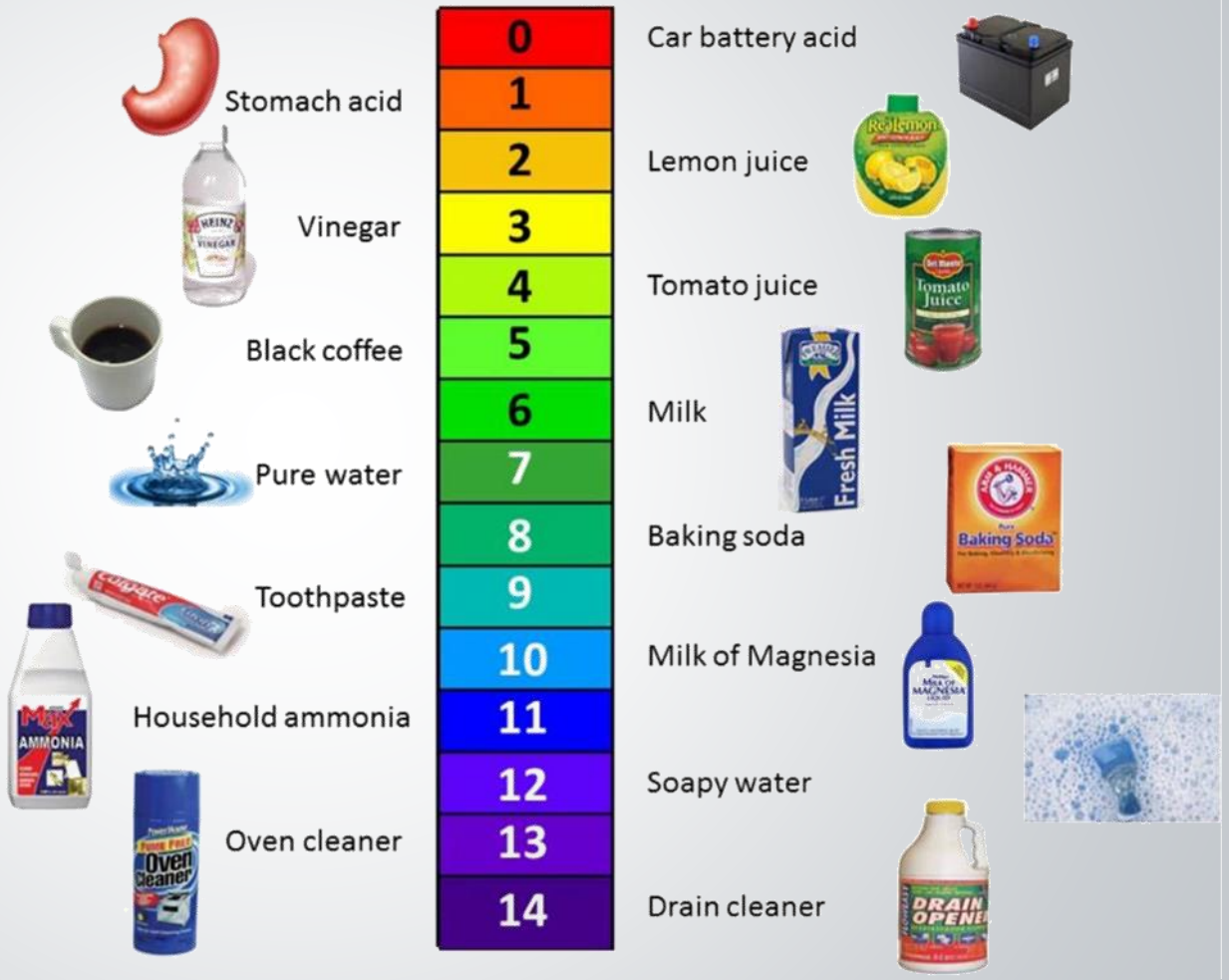
Henderson – Hasselbalch equation

Buffer capacity, β

2.1-1 pH for different solutions

pH testing strips

pH meter



pH of various substances

The *p*H scale

pH approximates to negative logarithm of the $[H^+]$.

Approximate definition of pH

$$pH \approx -\log[H^+]$$

When we measure pH with the pH meter, we measure the negative log of the H^+ activity, not its concentration.

The $[H^+]$ in pure water is 1.0×10^{-7} M, so the pH of water is

$$-\log(1.0 \times 10^{-7}) = \mathbf{7.00}$$

The *p*H range generally:

Neutral

acidic alkaline

*p*H= 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

The *p*H scale

$$pH \approx -\log[H^+]$$

$$p(OH) \approx -\log[OH^-]$$

$pH + p(OH) = -\log(Kw) =$ **14.00** at 25.0 °C

A solution is acidic if $[H^+] > [OH^-]$ pH < 7 at 25.0 °C

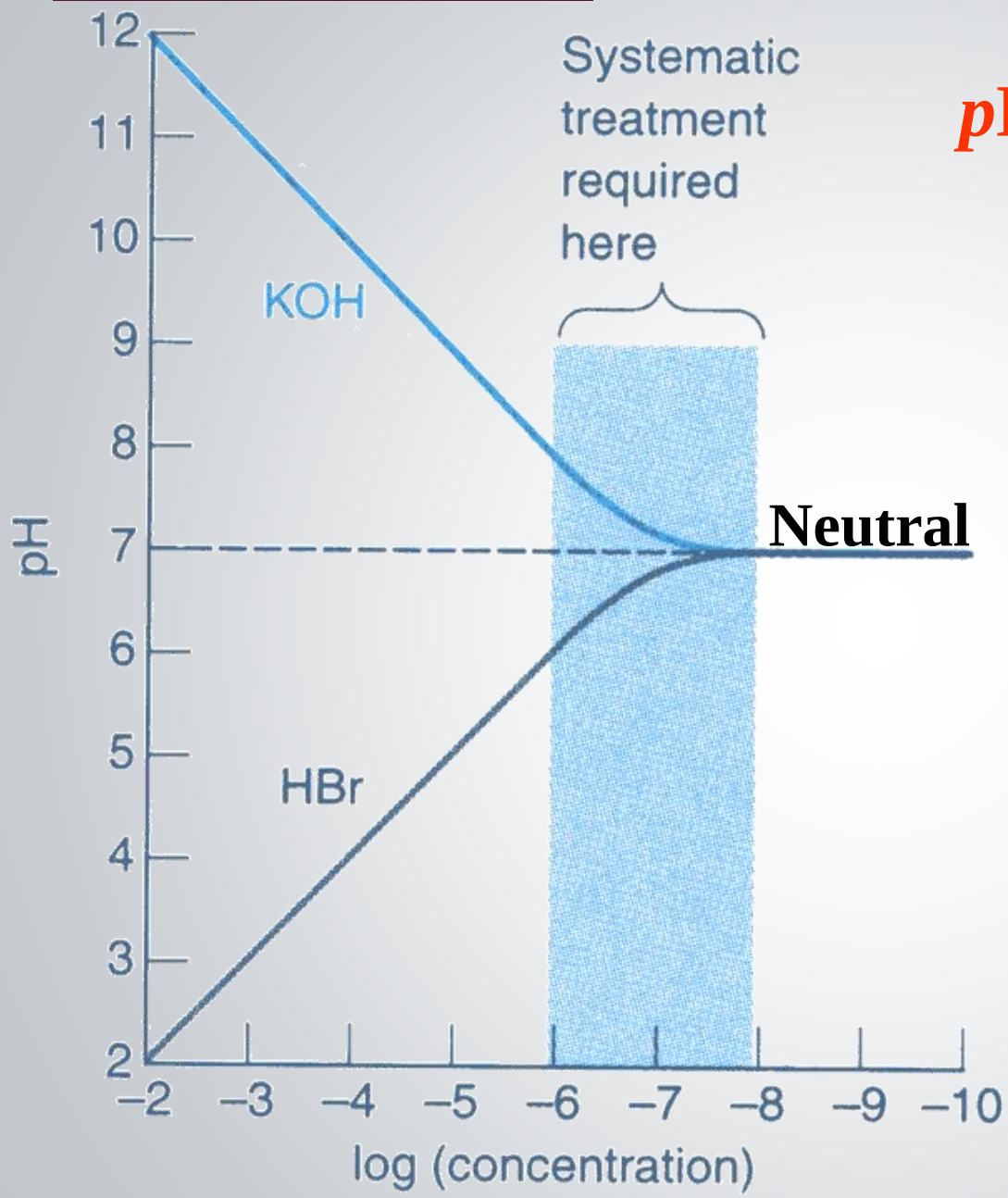
A solution is basic if $[H^+] < [OH^-]$ pH > 7 at 25.0 °C

Neutral

acidic ↓ alkaline

*p*H= 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

*p*H for strong acid/base



- For practical [C] $>10^{-6}$ M, pH or pOH is directly from the formal [C] of acid or base
- When [C] is near 10^{-7} M, we use systematic treatment of equilibrium to compute pH
- As [C] $<10^{-8}$ M, the pH is 7

Calculate pH as a function of [C] of strong acid & base

[C]	10^{-2}	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-7}	10^{-8}	10^{-9}	10^{-10}	M

Systematic treatment with *p*H for strong acid/base

f is the [C] of acid, near 10^{-7} M

$$Ka = \frac{(x+y)x}{f-x}$$

$$Kw = (x+y)y$$

$$(Kw+Ka)\,(f-x) = -x^2y - xy^2 + x^2 + fy^2 + (f+1)xy$$

HA	$\overset{K_a}{\rightleftharpoons}$	H^+	$+ A^-$
f		0	0
$f\text{-}x$		x	x
H_2O	$\overset{K_w}{\rightleftharpoons}$	H^+	$+ OH^-$
		y	y

Exact value requires solution of a third order equation.

$$Kw = (f + y)\,y$$

$Ka \rightarrow \infty$

$x \rightarrow f$

Solution using approximations

Systematic treatment with *p*H for strong acid/base

$$Kw = (f + y)\,y$$

$$y^2 + fy - Kw = 0$$

$$\begin{array}{ccccc} HA & \overset{K_a}{\rightleftharpoons} & H^+ & + & A^- \\ f & & - & & - \\ 0 & & f & & f \end{array}$$

$$\begin{array}{ccccc} H_2O & \overset{K_w}{\rightleftharpoons} & H^+ & + & OH^- \\ & & y & & y \end{array}$$

f is the [C] of acid, near 10^{-7} M

$y = [H^+]$ contribution from water

$$y = \frac{-f + \sqrt{f^2 + 4Kw}}{2}$$

Strong acid

$$[H^+] = x + y \approx f + y = \frac{f + \sqrt{f^2 + 4Kw}}{2}$$

Strong base

$$[H^+] = y = \frac{-f + \sqrt{f^2 + 4Kw}}{2}$$

Strong acids & bases in H_2O

- What is the pH of a 0.10 M HBr solution?

Since HBr is a strong acid we can assume that it dissociate 100%, so $Ka \rightarrow \infty$

$$[H^+] = x + y \approx f + y = \frac{f + \sqrt{f^2 + 4Kw}}{2}$$

$$[H^+] = \frac{0.10 + \sqrt{0.10^2 + 4 \times 10^{-14}}}{2} = 0.10 \text{ M}$$

$$pH = -\log[H^+] = -\log 0.10 = 1.00$$

- What is the pH of a 1.0×10^{-7} M HBr solution?

pH = 6.79

Strong acids & bases in H_2O

- What is the pH of 1.0×10^{-8} M NaOH solution?

Since NaOH is a strong base, the formula is

$$[H^+] = y = \frac{-f + \sqrt{f^2 + 4Kw}}{2}$$

$$[H^+] = \frac{-1.0\times10^{-8} + \sqrt{(1.0\times10^{-8})^2 + 4\times10^{-14}}}{2} = 9.5\times10^{-8}\ \text{M}$$

$$pH = -\log[H^+] = -\log 9.5\times10^{-8} = 7.02$$

Strong/weak acids & bases in H_2O

- Strong acid is dissociated completely, so $Ka \rightarrow \infty$
- Weak acid is only partially dissociated in H_2O, so Ka is "small"

Strong acids	Strong bases
HCl	LiOH
HBr	NaOH
HI	KOH
HNO_3	RbOH
$HClO_4$	CsOH
*H_2SO_4	

When either K_a or K_b is large, the acid or base is said to be strong. Common of them listed in Table

HBr Ka= $10^{5.8}$

Acetic acid Ka=$10^{-4,8}$

H_2SO_4, only the first proton ionization is complete.

Weak acids & bases in H_2O

$$HA \overset{K_a}{\rightleftharpoons} H^+ + A^-$$

f	-	-
f-x	x	x

f is the [C] of acid

when $f\,K_a > 10\,K_w$ (10^{-13}), the hydrolysis of H_2O can be neglected

$$Ka = \frac{(x+y)x}{f-x} \approx \boxed{\frac{x^2}{f-x}}$$

$$x^2 + Ka \cdot x - Ka \cdot f = 0$$

Fraction of dissociation

$$\alpha(A^-) = \frac{[A^-]}{[A^-]+[HA]} = \frac{x}{f}$$

$$\boxed{x = [H^+] = \frac{-Ka + \sqrt{Ka^2 + 4fKa}}{2}}$$

What is the pH of a 0.1 M acetic acid solution?

Ka of CH_3COOH is 1.75×10^{-5}

$[H^+] = 1.32 \times 10^{-3}$ M; pH is 2.88;

$\alpha(A^-) = 1.32\%$; $\alpha(HA) = 98.68\%$

Summery: Weak acids

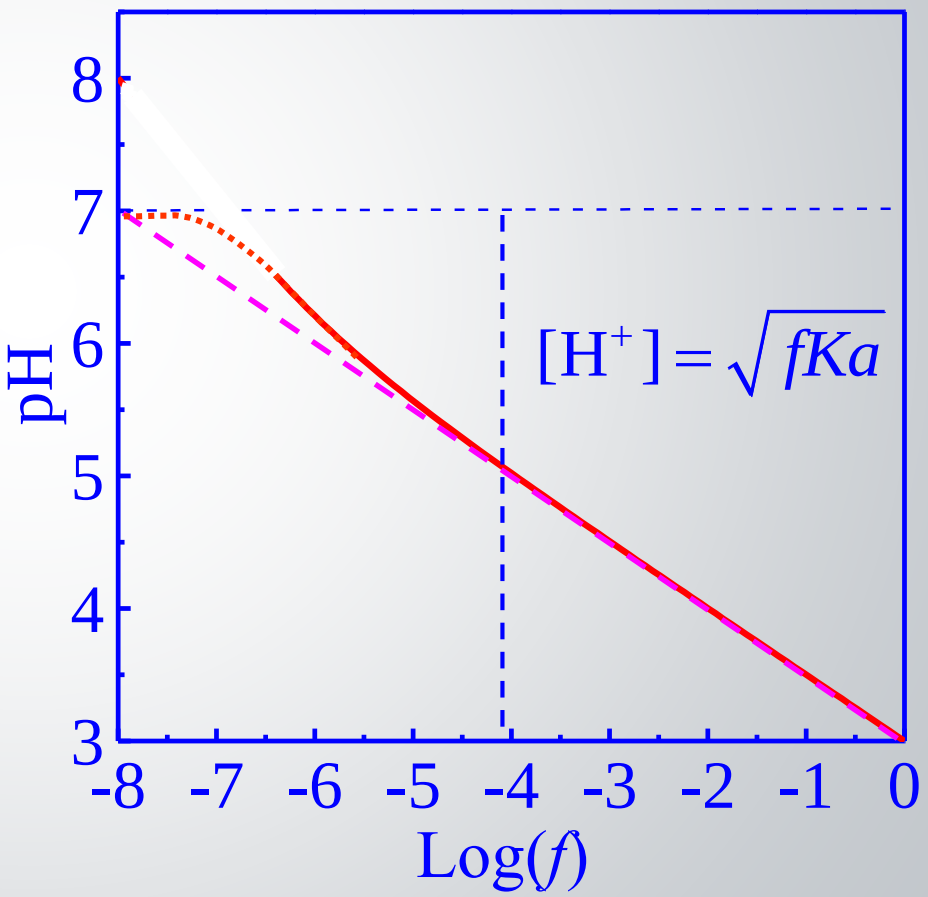
$$[H^+]^3 + K_a[H^+]^2 - (K_a f + K_w)[H^+] - K_a K_w = 0$$

Numerical solution of the 3rd order eq. ·····

$$[H^+] = \frac{-Ka + \sqrt{Ka^2 + 4fKa}}{2}$$

When $\frac{f}{K_a} \geq 100$, weak acid dissociation can be neglected

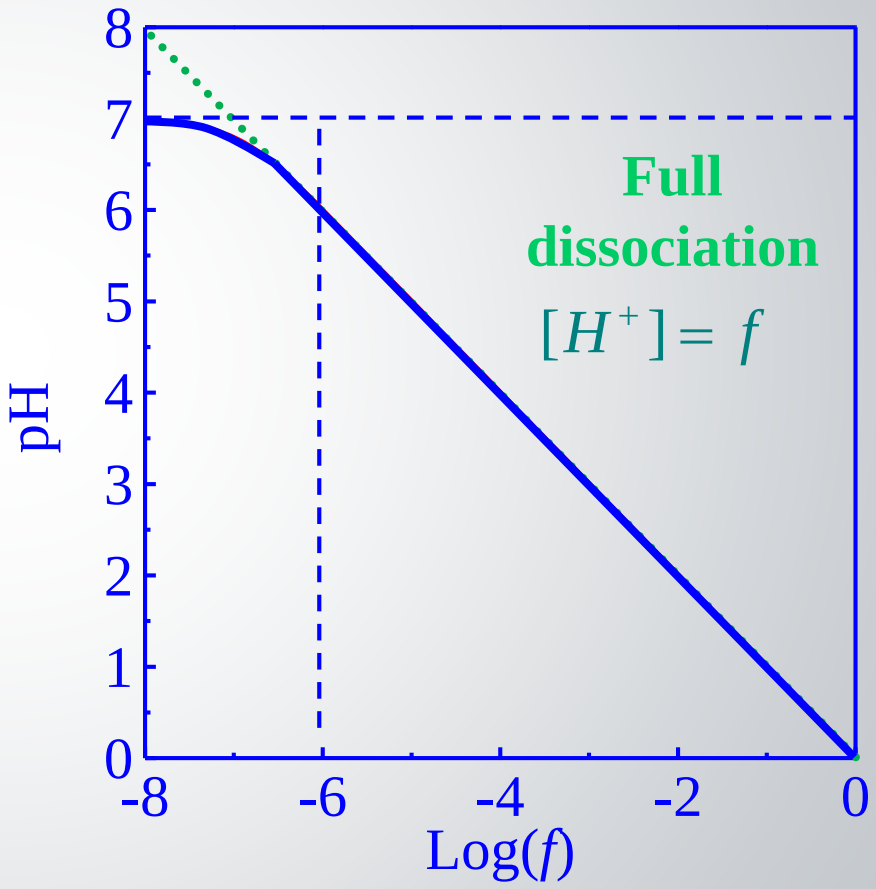
$$Ka \approx \frac{x^2}{f - x} \approx \frac{x^2}{f} \Rightarrow \boxed{[H^+] = \sqrt{fKa}}$$



Summery: strong acids

Numerical solution of the 3rd order eq. ———

$$[H^+] = \frac{f + \sqrt{f^2 + 4Kw}}{2}$$

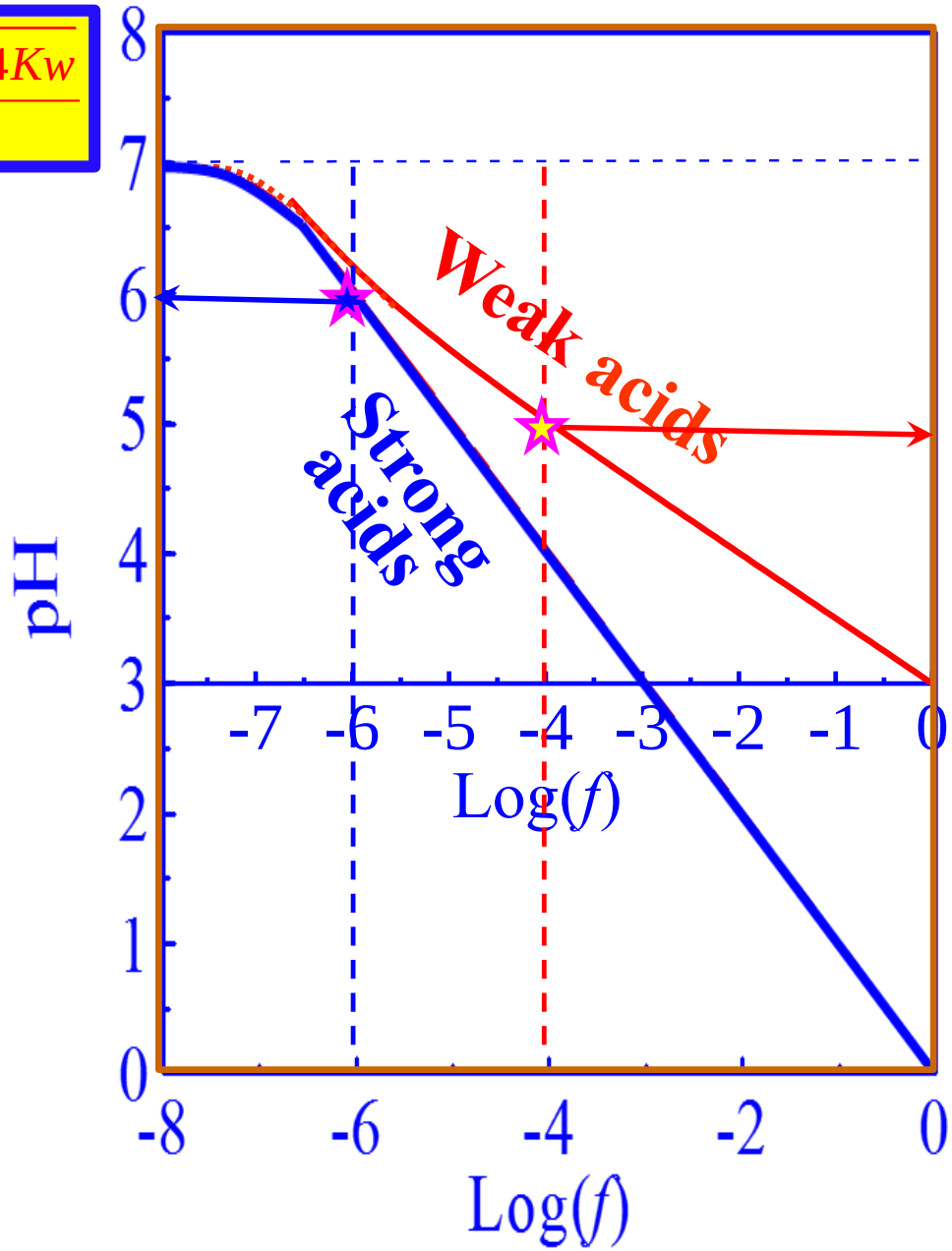


* Check the [Acid] *f* firstly

$$[H^+] = \frac{f + \sqrt{f^2 + 4Kw}}{2}$$

$f \geq 10^{-6}$ M

$[H^+] = f$



$$[H^+] = \frac{-Ka + \sqrt{Ka^2 + 4fKa}}{2}$$

$f \geq 100\ K_a$

$[H^+] = \sqrt{fKa}$

2.1-2 Buffer solutions

Definition: a solution that only slightly changes its *p*H upon addition of acid, base or dilution.

Function: to control the pH at a certain range, which is important for the chemical/biological reactions.

Preparation: (a) weak acid + its conjugated base
(b) weak base + its conjugated acid

Buffer solutions

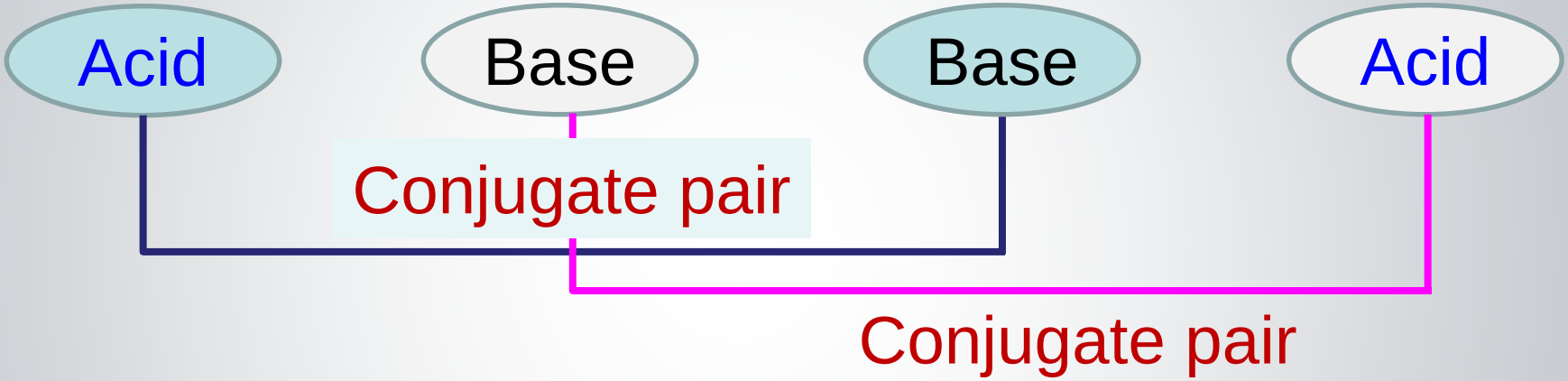
Examples:

HOCl	+	NaOCl
HA		A^-
weak acid		conjugated base

$C(CH_2OH)_3NH_3^+Cl^-$	+	$C(CH_2OH)_3\ NH_2$
BH^+	+	B
conjugated acid		weak base

Conjugate Acids and Bases

$$CH_3COOH + CH_3NH_2 \leftrightarrows CH_3COO^- + CH_3NH_3^+$$



Brönsted-Lowry Acids and Bases
(proton donors and acceptors)

Acid — which can donate a proton and become a base

Base — which can accept a proton to make a acid

Calculation *p*H of buffer
as a function of ingredients' concentration

Henderson – Hasselbalch equation

$$Ka = \frac{[H^+][A^-]}{[HA]}$$

$$HA \overset{K_a}{\rightleftharpoons} H^+ + A^-$$

$$\log Ka = \log[H^+] + \log\frac{[A^-]}{[HA]}$$

Approximation:

$[HA]_o$, $[A^-]_o$ >> $[H^+]$, $[OH^-]$

$$\boxed{pH = pKa + \log\frac{[A^-]}{[HA]}}$$

pH depends on concentration ratio, it don't change too much by dilution.

Calculation of buffer *p*H
as a function of ingredients' concentration

Henderson – Hasselbalch equation

$$pH = pKa + \log\frac{[A^-]}{[HA]}$$

When $[A^-] = [HA]$

$pH = pKa$

For every power of 10 change in the ratio $[A^-]/[HA]$, the pH changes only by one unit.

Effect of $[A^-]/[HA]$ on pH

$[A^-]/[HA]$	pH
100:1	p*Ka*+2
10:1	p*Ka*+1
1:1	p*Ka*
1:10	p*Ka*-1
1:100	p*Ka*-2

Buffer capacity, β

A quantity that represents the resistance of *p*H upon addition of strong acid/base.

$$\beta = \frac{dC_b}{dp\mathrm{H}} = -\frac{dC_a}{dp\mathrm{H}}$$

Where C_a and C_b *are the* moles of strong acid/base per 1 L to produce a unit change in pH.

large $\beta \rightarrow$ More resistant pH change

Calculation of buffer capacity

1L buffer contains
- C_b moles NaOH
- C_a moles HCl
- C moles weak acid ($HA + A^-$)

Equilibrium constants:

$$Ka = \frac{[H^+][A^-]}{[HA]}$$

$$Kw = [H^+][OH^-]$$

Mass conservation:

$$[Na^+] = C_b$$
$$[Cl^-] = C_a$$
$$[HA] + [A^-] = C$$

Charge conservation: $[H^+] + [Na^+] = [OH^-] + [Cl^-] + [A^-]$

$$Ka = \frac{[H^+][A^-]}{[HA]} \rightarrow [A^-] = \frac{Ka[HA]}{[H^+]} \rightarrow [A^-] = \frac{Ka(C - [A^-])}{[H^+]}$$

$$Kw = [H^+][OH^-]$$

$[Na^+] = C_b$

$[Cl^-] = C_a$

$[HA] + [A^-] = C \rightarrow [HA] = C - [A^-]$

$$[H^+] + [Na^+] = [OH^-] + [Cl^-] + [A^-]$$

$$[H^+] + C_b = Kw/[H^+] + C_a + [A^-]$$

$$[A^-] = \frac{CKa}{Ka + [H^+]}$$

$$C_b = C_a + \frac{Kw}{[H^+]} - [H^+] + \frac{CKa}{Ka + [H^+]}$$

$$C_b = C_a + \frac{Kw}{[\mathrm{H}^+]} - [\mathrm{H}^+] + \frac{CKa}{Ka + [\mathrm{H}^+]}$$

$$\beta = \frac{dC_b}{dp\mathrm{H}} = \frac{dC_b}{d[\mathrm{H}^+]}\frac{d[\mathrm{H}^+]}{dp\mathrm{H}} = -2.3[\mathrm{H}^+]\frac{dC_b}{d[\mathrm{H}^+]}$$

$$p\mathrm{H} = -\log_{10}[\mathrm{H}^+] = -\frac{1}{2.3}\ln[\mathrm{H}^+]$$

$$\frac{dp\mathrm{H}}{d[\mathrm{H}^+]} = -\frac{1}{2.3[\mathrm{H}^+]} \Rightarrow \frac{d[\mathrm{H}^+]}{dp\mathrm{H}} = -2.3[\mathrm{H}^+]$$

$$\beta = 2.3\left(\frac{Kw}{[\mathrm{H}^+]} + [\mathrm{H}^+] + \frac{CKa[\mathrm{H}^+]}{(Ka + [\mathrm{H}^+])^2}\right)$$

Buffer capacity, β

$$HA \rightleftharpoons H^+ + A^-$$

Ka of CH_3COOH is 1.75×10^{-5}

Equilibrium constant

$$K_a = \frac{[H^+][A^-]}{[HA]} \rightarrow \frac{[A^-]}{[HA]} = \frac{K_a}{[H^+]}$$

Fraction of dissociation

$$\alpha(A^-) = \frac{[A^-]}{[A^-]+[HA]} = \frac{K_a}{K_a+[H^+]}$$

$$\alpha(HA) = \frac{[HA]}{[A^-]+[HA]} = \frac{[H^+]}{K_a+[H^+]}$$

$$\alpha(HA) \times \alpha(A^-) = \frac{K_a[H^+]}{(K_a+[H^+])^2}$$

When pH = pK_a, $\alpha(HA) = \alpha(A^-) = 0.5$

$$\beta = 2.3\left([H^+] + \frac{CKa[H^+]}{(Ka+[H^+])^2} + [OH^-]\right)$$

$$\beta_{max} = 2.3\left(Ka + \frac{CKa^2}{4Ka^2} + \frac{Kw}{Ka}\right)$$

$$\beta = 2.3\left(Ka + \frac{C}{4} + \frac{Kw}{Ka}\right)$$

$$\beta = 2.3\left(1.75\times10^{-5} + 0.025C + 5.71\times10^{-10}\right)$$

$$\beta_{max} \approx 2.3 \times 0.025C = 0.0575C$$

$$\beta = 2.3C\times\alpha(HA)\times\alpha(A^-)$$

$$\Rightarrow \beta_{max.} = 0.575\,C$$

Buffer capacity, β

$$\beta_{HA-A^-} = 2.3 \cdot C \cdot \alpha_{HA} \cdot \alpha_{A^-}$$

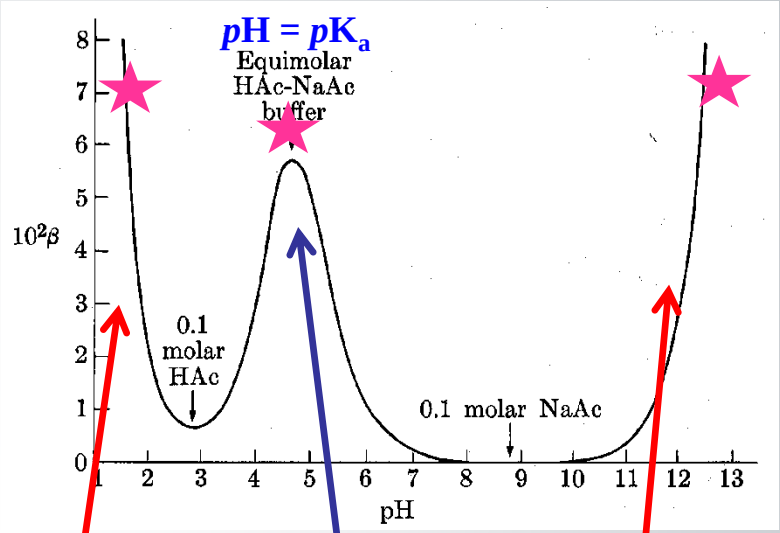
pH=pK_a	α_{HA}= 0.5, α_{A^-}= 0.5	β= β_{max} β_{max}=0.575C
pH=pK_a-1	α_{HA}= 0.9, α_{A^-}= 0.1	β= 0.36β_{max} =0.207C
pH=pK_a+1	α_{HA}= 0.1, α_{A^-}= 0.9	β= 0.36β_{max} =0.207C
pH=pK_a - 2	α_{HA}= 0.99, α_{A^-}= 0.01	β= 0.04β_{max} =0.023C
pH=pK_a + 2	α_{HA}= 0.01, α_{A^-}= 0.99	β= 0.04β_{max} =0.023C

Preparing buffer at maximum capacity

$$F = \frac{CKa[H^+]}{(Ka+[H^+])^2}$$

$$\frac{dF}{d[H^+]} = 0 = \frac{CKa\,(Ka-[H^+])}{(Ka+[H^+])^2}$$

When $[H^+] = Ka$, F is maximal



3rd approximation of Brönsted eq:

$$pH = pKa + \lg\frac{[A^-]_0}{[HA]_0}$$

$$\beta = 2.3\left([H^+] + \frac{CKa[H^+]}{(Ka+[H^+])^2} + [OH^-]\right)$$

When $[A^-]_0 = [HA]_0$, then $pH = pKa$ and F is at maximum

Maximum buffer capacity when $[salt]_0 = [acid]_0$

* Calculation of $[H^+]$ for monoacid and buffer

- **Strong acid**

$$[H^+] = \frac{f + \sqrt{f^2 + 4Kw}}{2}$$

If $f \geq 10^{-6}$ M, $[H^+] = f$

- **Weak acid**

$$[H^+] = \frac{-Ka + \sqrt{Ka^2 + 4fKa}}{2}$$

when $f\,Ka > 10$ Kw

If $f \geq 100\,K_a$, $[H^+] = \sqrt{fKa}$

- **Buffer pH (HA-A$^-$)**

$$pH = pKa + \log\frac{[A^-]}{[HA]}$$

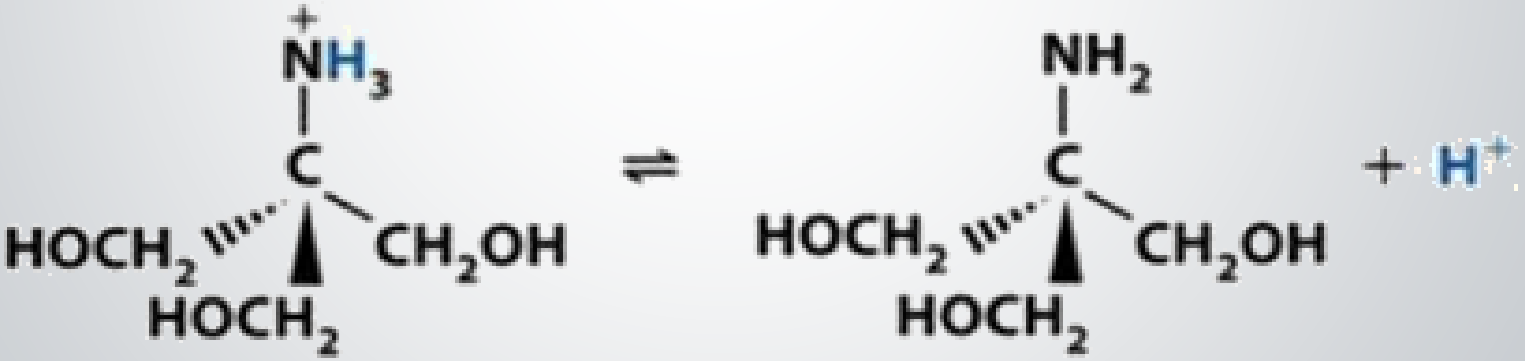
$$HA \overset{K_a}{\rightleftharpoons} H^+ + A^-$$

$$A^- + H_2O \overset{K_b}{\rightleftharpoons} HA + OH^-$$

$$K_aK_b = \frac{(H^+)\cdot\cancel{(A^-)}}{\cancel{(HA)}}\cdot\frac{\cancel{(HA)}\cdot(OH^-)}{\cancel{(A^-)}} = K_w$$

Calculation of buffer *p*H

- A widely used buffer is “tris”(hydroxymethyl-aminomethane, B), p*Ka*=8.072 for *its* conjugate acid (BH^+)
- What is the pH of a 1.00 L aqueous solution with 12.43 g B of tris (FM 121.135) plus 4.67 g of tris hydrochloride BH^+ (FM 157.596) ?



BH⁺
(pK_a=8.072)

B
(This form is “tris”)

Calculation of buffer *p*H

- What is pH of **1.00 L** with 12.43 g of B (FM 121.135) plus 4.67 g of BH^+ (FM 157.596) ? p*Ka*=8.072

The [B] and [BH^+] are

$$[B] = \frac{12.43\ g/L}{121.135\ g/mol} = 0.1026\ M$$

$$[BH^+] = \frac{4.67\ g/L}{157.596\ g/mol} = 0.0296\ M$$

Henderson – Hasselbalch equation

$$pH = pKa + \lg\frac{[A^-]}{[HA]} = 8.072 + \log\frac{0.1026}{0.0296} = 8.61$$

How about adding 12.0 mL of 1.00 M HCl?

Effect of adding acid to a buffer

What is the pH of 12.0 mL of 1.00 M HCl to the 1 L solution of tris with 0.1026 M of B and 0.0296 M of BH^+? pK_a =8.072

	B	+	H^+	→	BH^+
	Tris		From HCl		
Initial	0.1014 M		0.0119 M		0.0292 M

$$[B] = \frac{0.1026\frac{mol}{L}\times 1L}{1\ L + 12.0\ mL} = 0.1014\ M$$

$$[BH^+] = \frac{0.0296\ mol/L\times 1L}{1\ L + 12.0\ mL} = 0.0292\ M$$

$$[HCl] = \frac{1.00\ \frac{mol}{L}\times 12.0mL}{1\ L + 12.0\ mL} = 0.0119\ M$$

Effect of adding acid to a buffer

What is the pH of 12.0 mL of 1.00 M HCl to the 1 L solution of tris with 0.1026 M of B and 0.0296 M of BH^+? p*Ka*=8.072

	B	$+$	H^+	$\rightarrow$	BH^+
	Tris		From HCl		
Initial	0.1014 M		0.0119 M		0.0292 M
Final	0.0895 M↓		0.000 M		0.0411 M↑

The H^+ from HCl would consume by B to produce BH^+.

$$pH = pKa + \lg\frac{[A^-]}{[HA]} = 8.072 + \log\frac{0.0895}{0.0411} = 8.41$$

Effect of adding acid to a buffer

$$B + H^+ \rightarrow BH^+$$

Tris from HCl

The strong acid is consumed by B, as long as the B is not used up, the $\frac{[A^-]}{[HA]}$ does not change much

$$pH = pKa + \lg\frac{[A^-]}{[HA]} = 8.072 + \log \mathbf{3.47} = 8.61$$

12.0 mL of 1.00 M HCl added, ΔpH = -0.20

$$pH = pKa + \lg\frac{[A^-]}{[HA]} = 8.072 + \log \mathbf{2.18} = 8.41$$

A buffer resists change in pH when a limited amount of strong acid or base is added. Why?